

PROBABILITY

→ Simple Questions

→ Use of Combinations in Theory of Probability

→ Theorems of Probability

Addition
Theorem

Multiplication
Theorem

Bayes'
Theorem

→ Conditional Probability

→ Bernoulli's Theorem in Theory of Probability

→ Mathematical Expectation

Classical or Mathematical Definition

$$P(A) = \frac{\text{Number of favourable cases}}{\text{Total Number of Equally likely cases}}$$

$$P(\bar{A}) = 1 - P(A)$$

Use of Combinations in Theory of Probability

$${}^nC_r = \frac{n!}{r! (n-r)!}$$

THEOREMS OF PROBABILITY

→ Addition Theorem

→ Multiplication Theorem

→ Bayes' Theorem

ADDITION THEOREM

Addition Theorem for Mutually Exclusive Events

$$P(A \text{ or } B) = P(A) + P(B)$$

$$P(A+B+C) = P(A) + P(B) + P(C)$$

Addition Theorem for Not Mutually Exclusive Events.

$$P(A \text{ or } B \text{ or both}) = P(A) + P(B) - P(AB)$$

$$\begin{aligned} P(\text{Either } A \text{ or } B \text{ or } C) &= P(A) + P(B) + P(C) \\ &\quad - P(AB) - P(BC) - P(AC) \\ &\quad + P(ABC) \end{aligned}$$

MULTIPLICATION THEOREM

Multiplication Theorem for Independent Events

$$P(AB) = P(A) \times P(B)$$

NOTE →

Question Type

Probability of happening of at least one event
in case of n independent events

Multiplication Theorem for Dependent Events

Conditional Probability

Use of Bernoulli's Theorem in Theory of Probability

$$P(x) = {}^n C_x p^x \cdot q^{n-x}$$

$$x = 1, 2, 3, \dots, n$$

$P(x)$ = Probability of x successes in n trials.

p = Probability of success.

q = Probability of failure.

n = Total no of Trials.

Mathematical Expectation

$$M.E = P \times M$$

P = Probability of happening

M = Money to be received on the
happening of the event.

Bayes' Theorem

$$P(A_1|B) = \frac{P(A_1) P(B|A_1)}{P(A_1) P(B|A_1) + P(A_2) P(B|A_2)}$$

A_1 and A_2 are mutually exclusive events

B be an event which can occur in combination with A_1 and A_2 .